

MonsoonIQ — Verification Report

Regime-aware post-processing of NWP rainfall · held-out period [2024, 2025] · 234 days · 12402 district-days · generated 2026-09-26 05:31 UTC

DATA PROVENANCE: REAL_IMD_GFS — Observations: IMD 0.25 deg daily gridded rainfall (Pai et al. 2014, IMD Pune). Forecasts: NOAA GFS 00 UTC runs (AWS Open Data noaa-gfs-bdp-pds), Day 1-5. District boundaries: DataMeet Census-2011 (CC BY 2.5 IN). GFS stands in for NCMRWF NCUM, which is not publicly downloadable. All skill figures below are measured against this archive and are **not** evidence of operational skill.

1 · Scorecard (held-out period)

Quantity	Raw NWP	Regime-agnostic	Regime-aware	Change vs raw	Change vs baseline
RMSE (mm/day)	14.745	13.083	13.058	-1.687	-0.025
Heavy-rain CSI (≥64.5 mm)	0.085	0.167	0.155	+0.0700 (significant)	-0.0121 (significant)
Heavy-rain ETS	0.074	0.150	0.140	+0.066	-0.011
Very heavy CSI (≥115.6 mm)	0.045	0.012	0.009	-0.037	-0.003

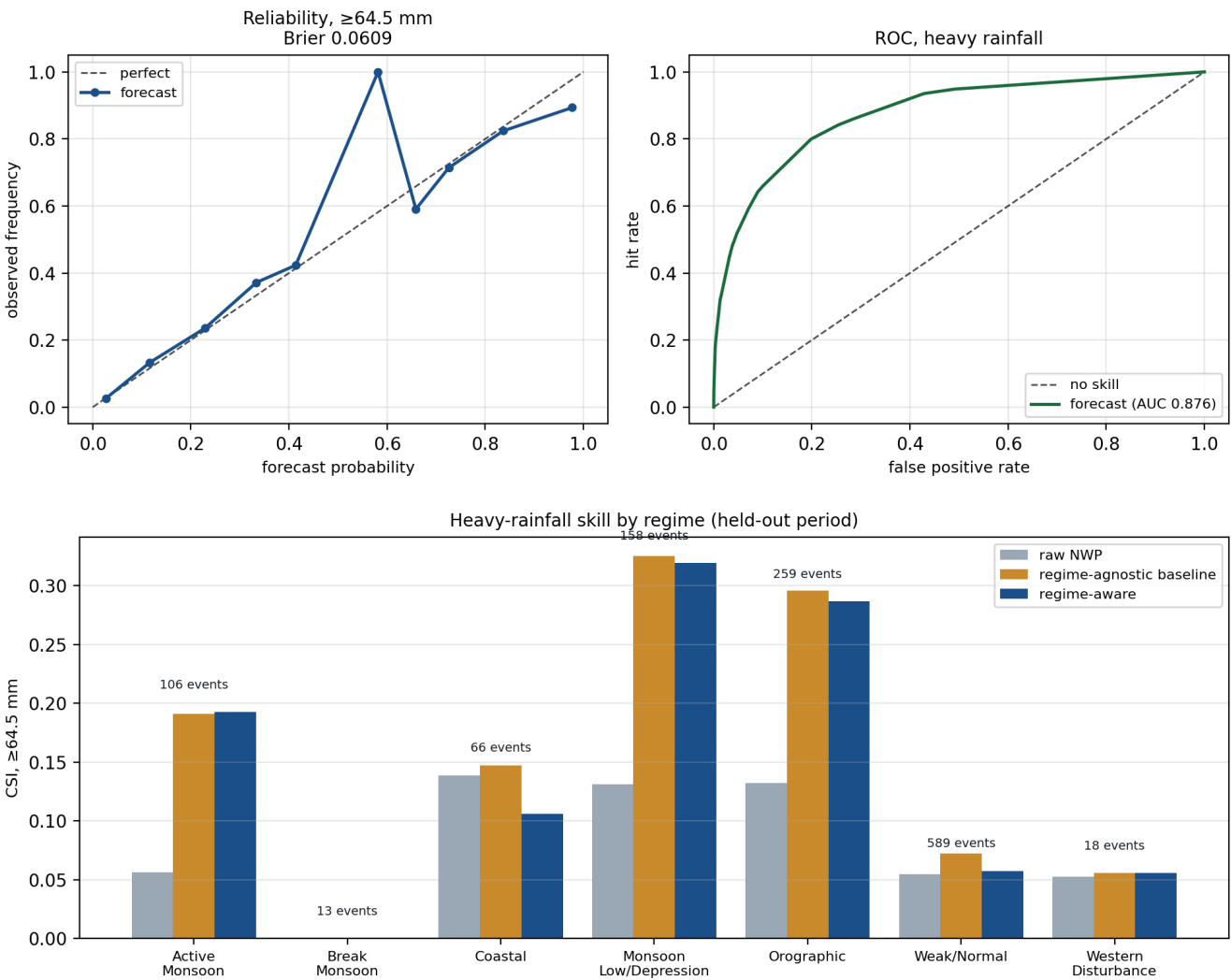
The regime-agnostic column is a gradient-boosted model fitted on the same predictors without regime information. It is the baseline that decides whether regime conditioning earns its complexity; the raw model alone does not.

2 · Heavy and very heavy rainfall, by regime

Regime	Days	Events (≥64.5)	≥115.6	Reliability	Raw CSI	Regime-agnostic CSI	Regime-aware CSI	Δ vs raw [95% CI]
Active Monsoon	583	106	42	reliable	0.056	0.191	0.193	+0.137 [+0.042, +0.241]
Break Monsoon	318	13	4	indicative	0.000	0.000	0.000	+0.000 [+0.000, +0.000]
Coastal	583	66	19	reliable	0.139	0.147	0.106	-0.033 [-0.112, +0.040]
Monsoon Low/Depression	1060	158	42	reliable	0.131	0.325	0.319	+0.188 [+0.110, +0.255]
Orographic	1219	259	110	reliable	0.132	0.296	0.287	+0.155 [+0.103, +0.210]
Weak/Normal	8533	589	120	reliable	0.055	0.072	0.058	+0.003 [-0.022, +0.026]
Western Disturbance	106	18	5	indicative	0.053	0.056	0.056	+0.003 [-0.100, +0.111]

Strata with fewer than 10 events are marked **insufficient**: their CSI is printed for completeness only and is not a scored result. Intervals are day-block bootstraps — whole calendar days are resampled, because 53 districts on one day are a single weather system.

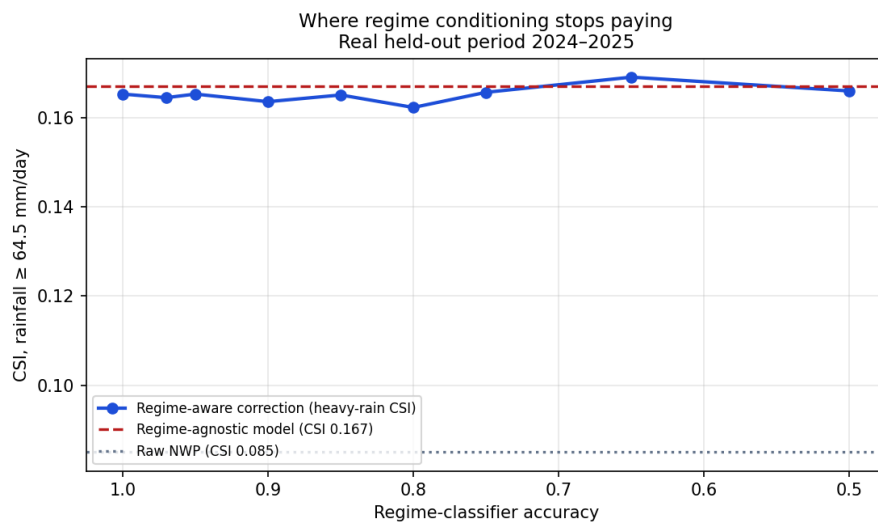
3 · Reliability, ROC and regime breakdown



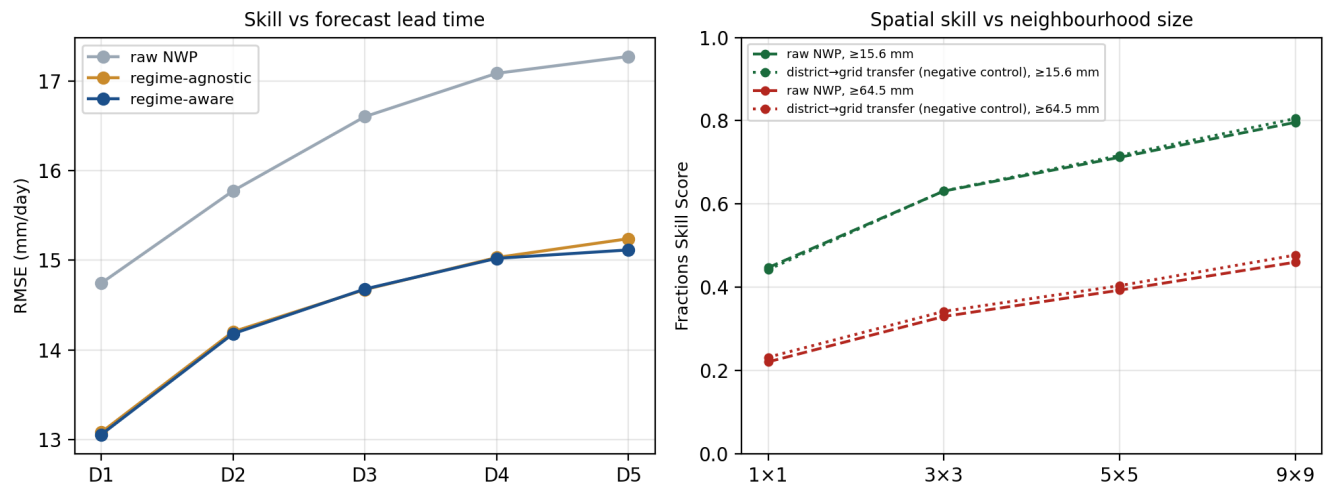
4 · What regime awareness is worth

Experiment	Result
Correction alone, uniform regime weights (CSI)	0.1671
Correction + classifier regime weights (CSI)	0.1550
Regime-agnostic model, same predictors (CSI)	0.1671
Gain attributed to regime awareness	-0.0121 [-0.0225, -0.0026] — significant
Classifier accuracy at which the gain vanishes	0.65

Uniform regime weights reproduce a 'correction with no regime information' system; the gap between it and the classifier blend is the actual value of regime awareness. If that gap is inside the interval, the regime layer is not yet earning its complexity and the honest fix is more training seasons or a better regime classifier - not a stronger claim.



5 · Lead time and spatial scale



Two grid paths are scored. `\*\_district\_transfer` extends district-scale multiplicative adjustments to cells by nearest-district assignment; `regime\_aware\_grid\_native` is a correction fitted on grid cells. Grid truth is IMD 0.25 deg rainfall regridded to 0.5 deg; the raw grid forecast is GFS.

Threshold	Neighbourhood	Raw NWP	Grid-native	District→grid (control)
≥15.6 mm	1×1 cells	0.447	—	0.442
≥15.6 mm	3×3 cells	0.631	—	0.631
≥15.6 mm	5×5 cells	0.712	—	0.716
≥15.6 mm	9×9 cells	0.796	—	0.805
≥64.5 mm	1×1 cells	0.220	—	0.231
≥64.5 mm	3×3 cells	0.330	—	0.342
≥64.5 mm	5×5 cells	0.393	—	0.404
≥64.5 mm	9×9 cells	0.460	—	0.477

6 • Claim check

Claim	Verdict	Evidence
Post-processing reduces RMSE relative to the raw model	SUPPORTED	Δ RMSE -1.655 mm/day, 95% CI [-2.206, -1.181]
Post-processing improves the heavy-rainfall categorical scores relative to the raw model	NOT SUPPORTED	Δ CSI +0.0700, 95% CI [+0.0465, +0.0982]
Regime conditioning beats a regime-agnostic model fitted to the same predictors (heavy-rainfall 5 deg grid)	NOT SUPPORTED	Δ CSI -0.0111, 95% CI [-0.025, +0.0026]
Regime conditioning beats a regime-agnostic model on RMSE	SUPPORTED	Δ RMSE -0.025 mm/day, 95% CI [-0.096, +0.053]

Intervals resample whole calendar days. Observations are IMD 0.25 deg gridded rainfall and the raw model is NOAA GFS (a public stand-in for NCMRWF NCUM), so these are real verification results for 2024-2025 over 53 districts - not an operational IMD evaluation.

7 • Reproducibility

python src/train.py → python src/evaluate.py → pytest tests/ → uvicorn src.api.main:app
Model manifest: artifacts/models/training_manifest.json (Gradient boosting (regime-agnostic), MonsoonIQ (regime-aware), Quantile mapping (regime-agnostic), Raw NWP)
Every figure and number in this report is regenerated from the raw archive by the commands above; no value is hand-entered.